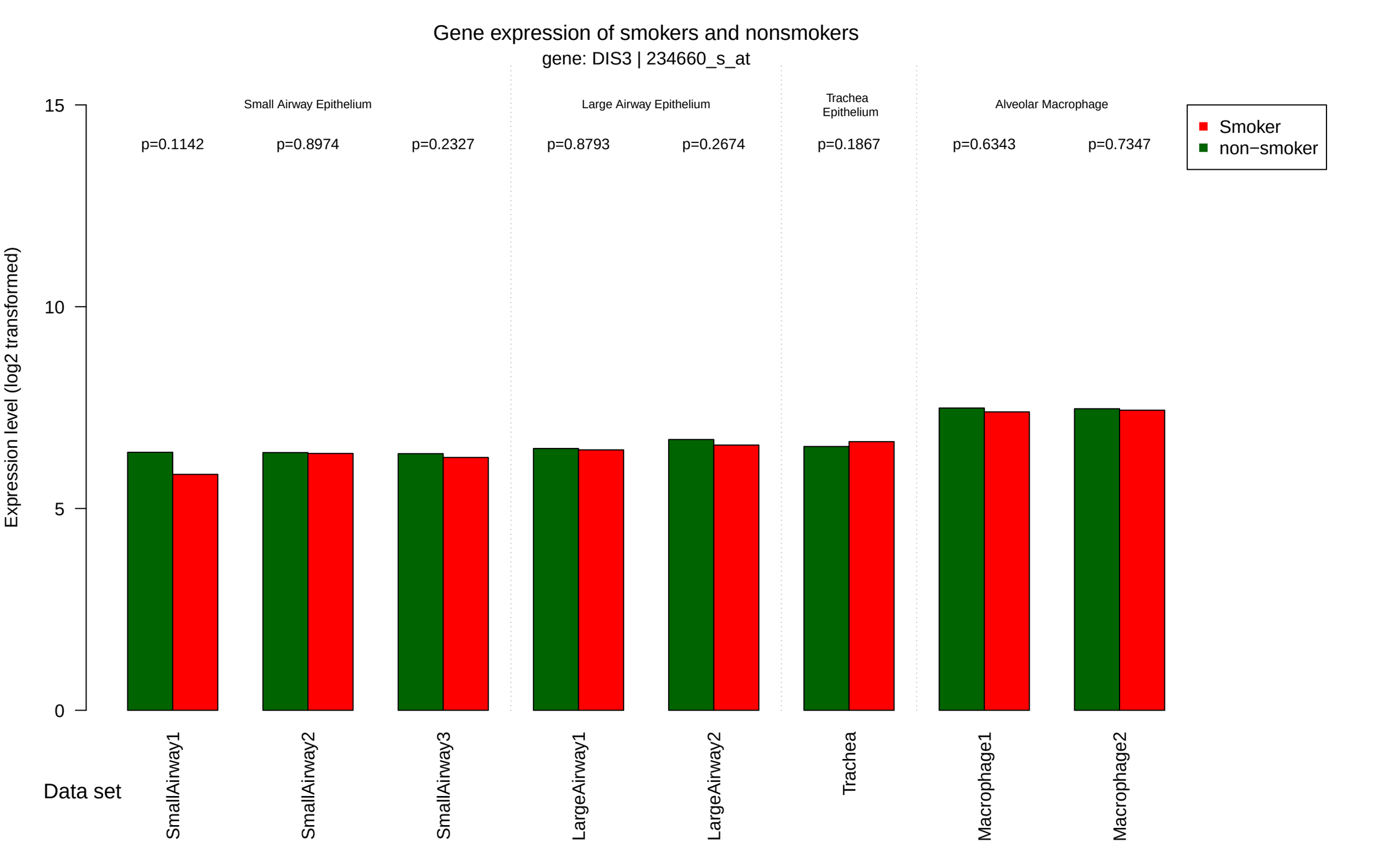
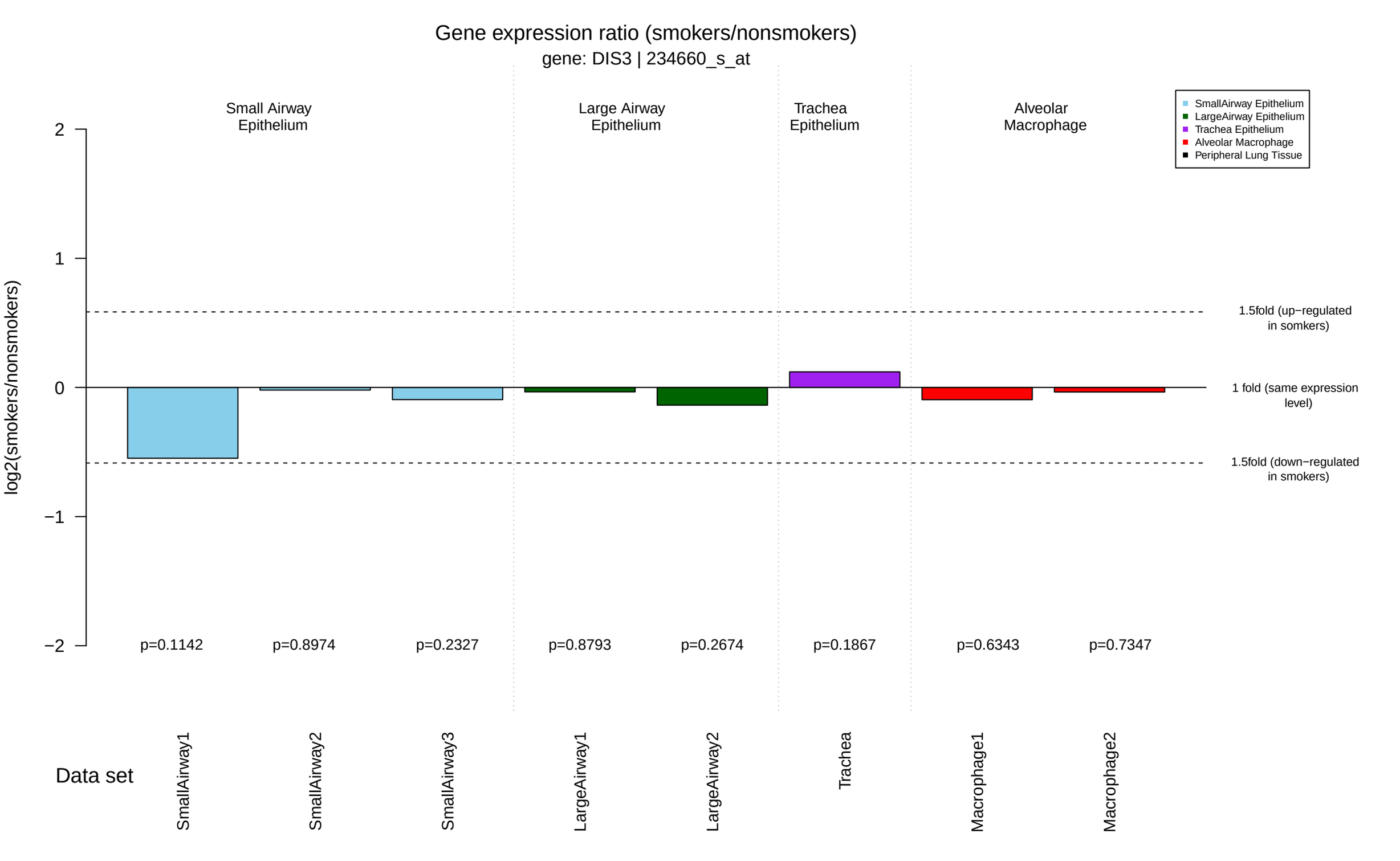
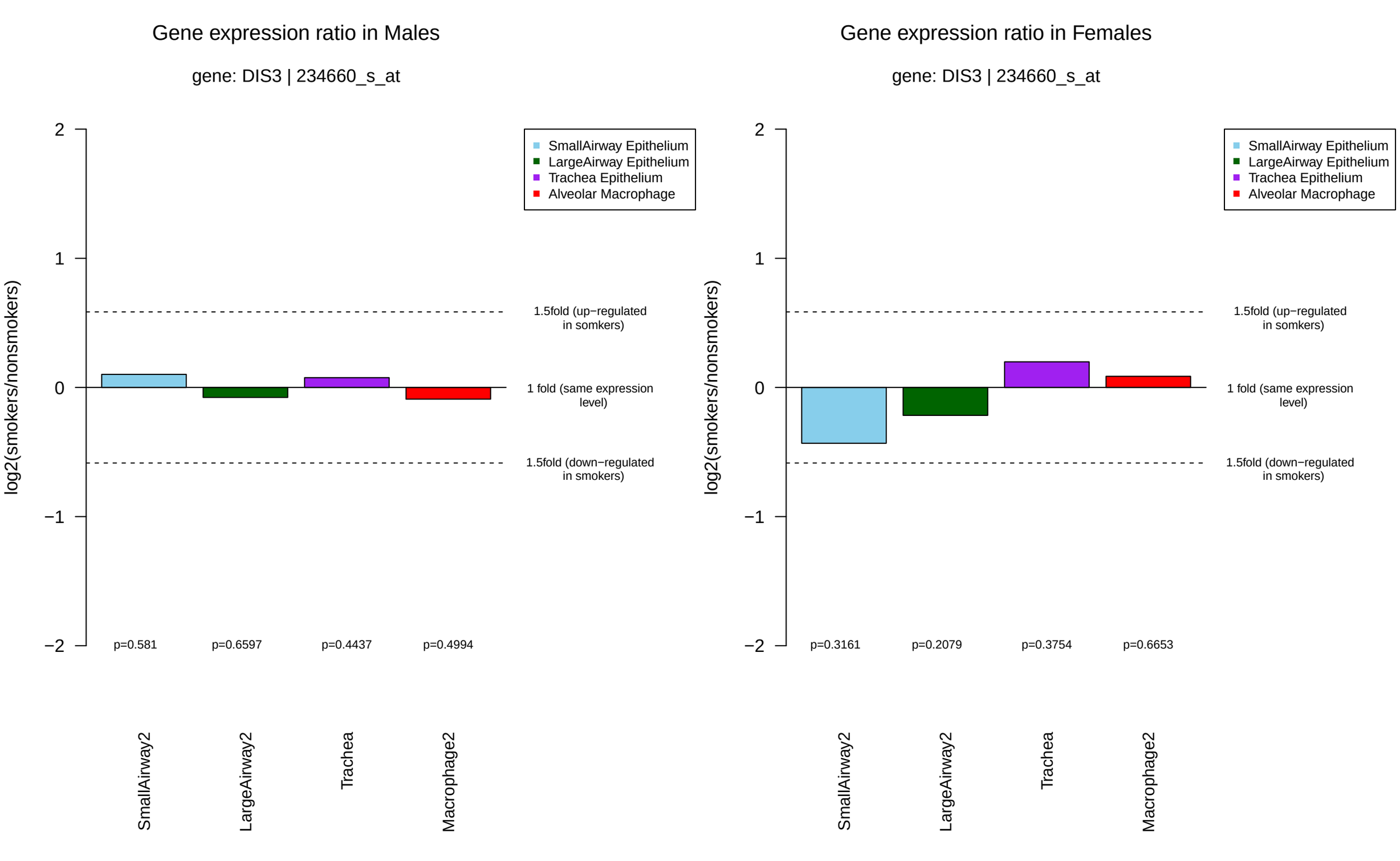
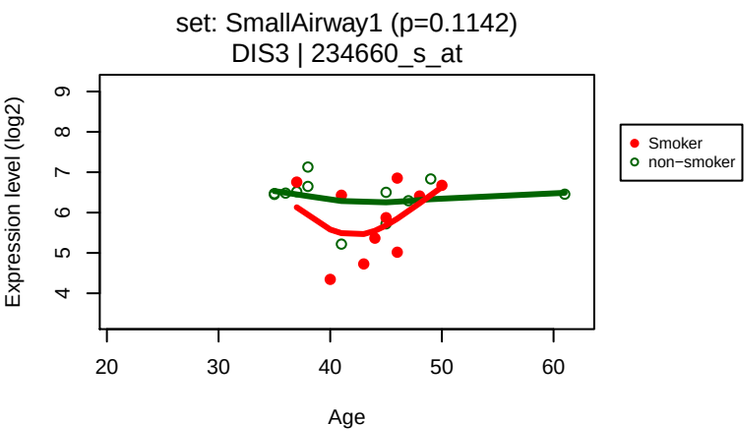
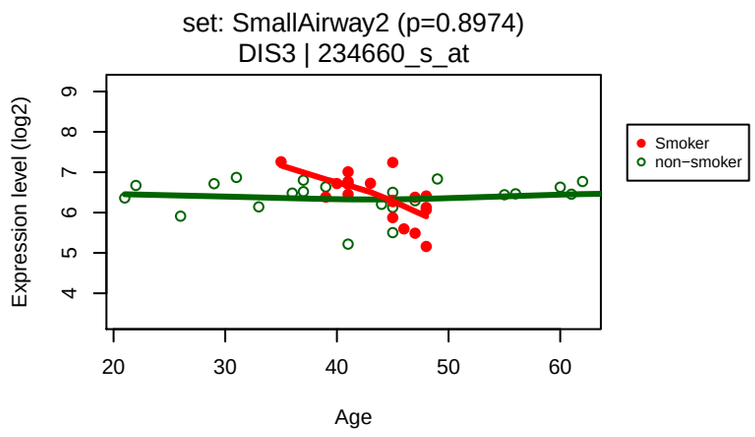
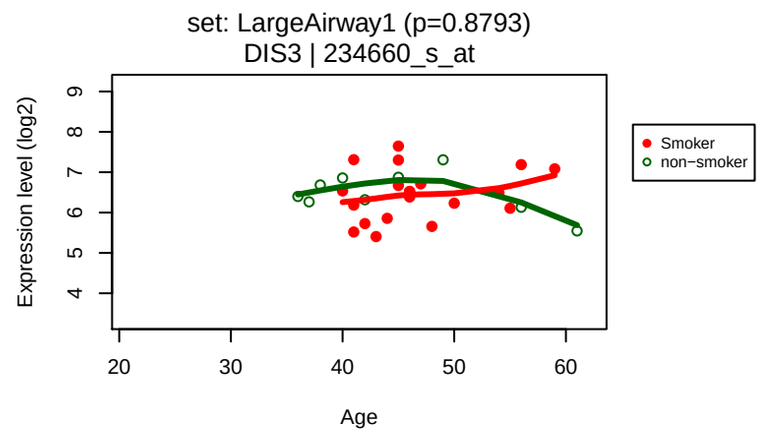

Gene expression of smokers and nonsmokers
gene: DIS3 | 234660_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1142
p=0.8974
p=0.2327
p=0.8793
p=0.2674
p=0.1867
p=0.6343
p=0.7347
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

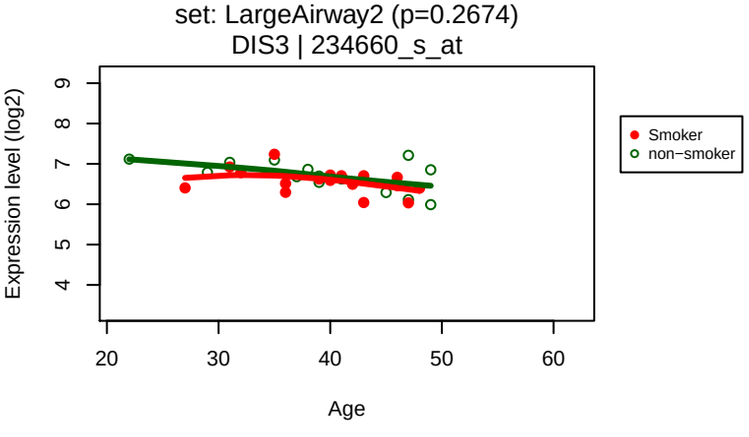

Gene expression ratio (smokers/nonsmokers)
gene: DIS3 | 234660_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1142
p=0.8974
p=0.2327
p=0.8793
p=0.2674
p=0.1867
p=0.6343
p=0.7347
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

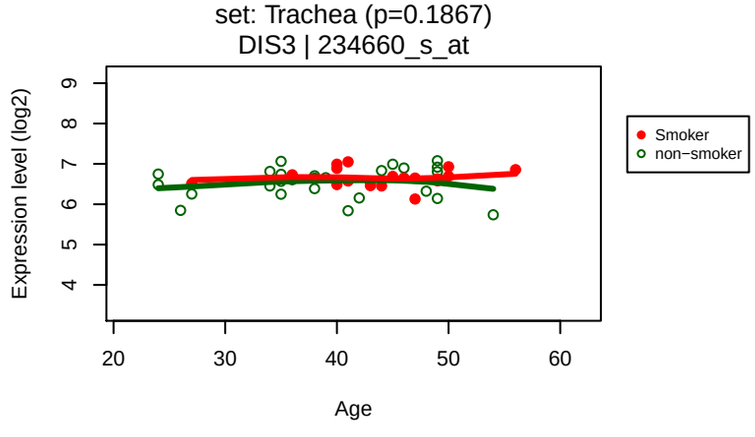

Gene expression ratio in Males
gene: DIS3 | 234660_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.581
p=0.6597
p=0.4437
p=0.4994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DIS3 | 234660_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3161
p=0.2079
p=0.3754
p=0.6653
SmallAirway2
LargeAirway2
Trachea
Macrophage2

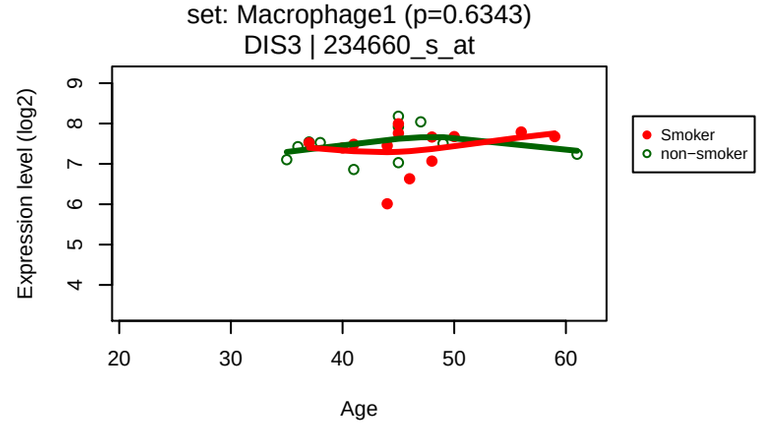

set: SmallAirway1 (p=0.1142)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker

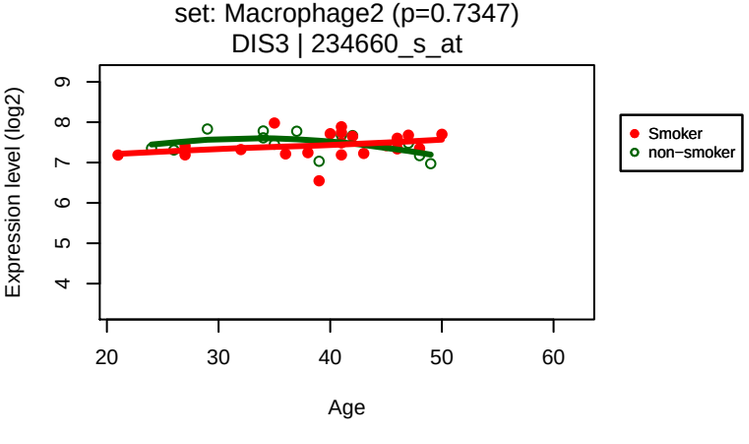

set: SmallAirway2 (p=0.8974)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker

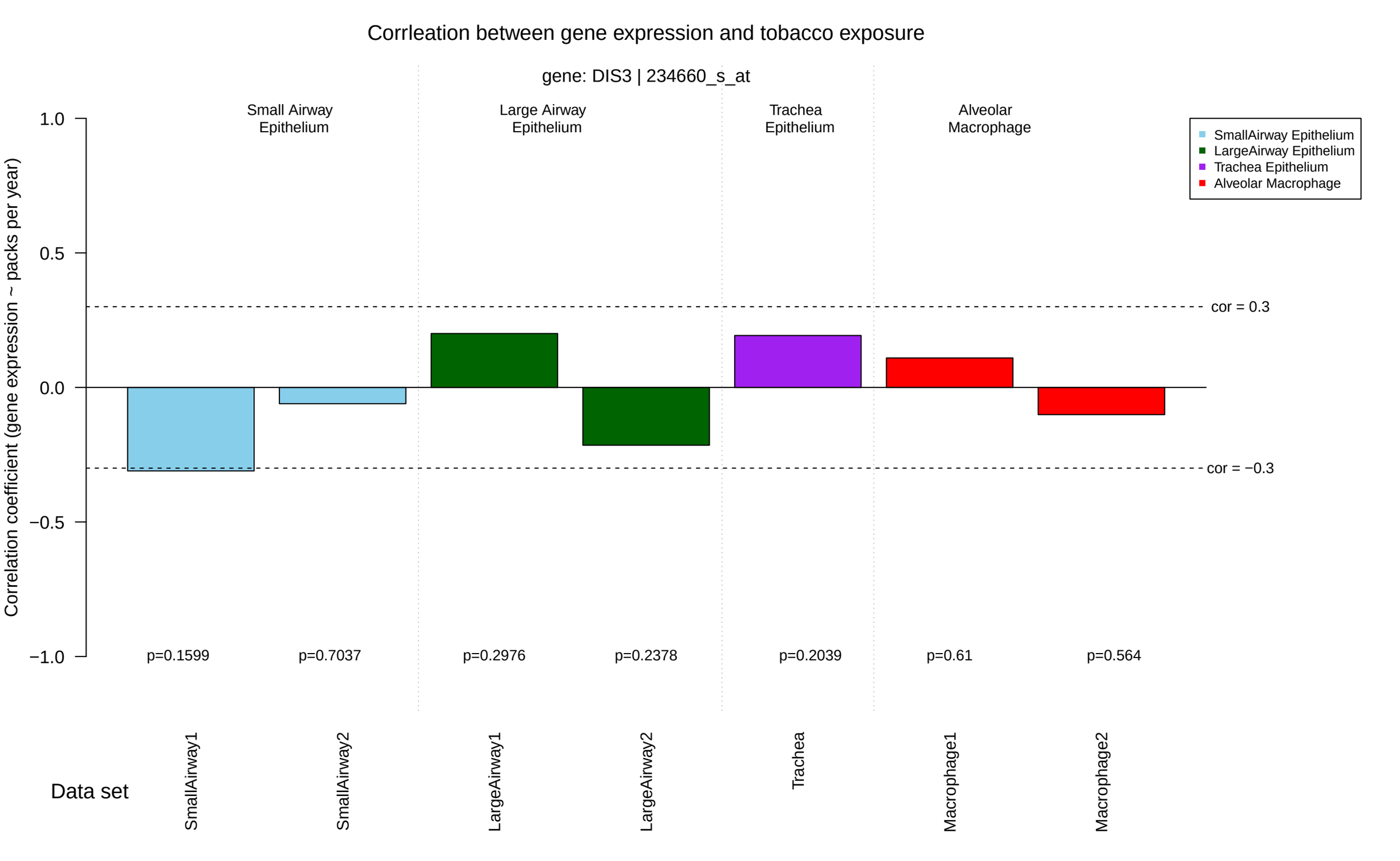

set: LargeAirway1 (p=0.8793)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2674)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1867)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6343)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7347)
DIS3 | 234660_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DIS3 | 234660_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1599
p=0.7037
p=0.2976
p=0.2378
p=0.2039
p=0.61
p=0.564
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2